

Marketing Analytics: Regression Analysis

Identifying Factors Associated with Customer Purchase Amount

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1. Introduction

Understanding what influences customer spending helps companies make better marketing decisions. This report analyzes the factors associated with customer purchase amount using multiple linear regression. Three models were developed and compared before selecting the final model based on statistical performance and diagnostic testing.

The following sections interpret the final model from a marketing perspective, discuss its practical implications, acknowledge its limitations, and suggest opportunities for future improvement.

2. Key Findings

Purchasing power is the strongest driver of customer spending.

The final regression model identified income as the strongest predictor of purchase amount. Even after accounting for customer engagement and promotional activity, higher-income customers consistently spent more than lower-income customers. This suggests that purchasing power influences the amount customers are willing or able to spend, regardless of how they interact with marketing activities. While engagement and promotions can encourage purchases, income appears to determine the overall spending capacity of the customer. From a marketing perspective, this finding supports customer segmentation based on spending potential rather than applying the same strategy to the entire customer base.

Customer engagement is strongly associated with higher spending.

Both website visits and the number of pages viewed were significant predictors of purchase amount, indicating that customers who interact more frequently with the website also tend to spend more. These two variables capture different aspects of engagement: returning to the website and actively exploring its content. Together, they suggest that customers who invest more time and attention before making a purchase are also more likely to spend more. From a marketing perspective, this highlights the value of creating digital experiences that encourage customers to return, browse products, and remain engaged throughout their purchasing journey.

Promotional campaigns increase purchase amount but do not replace customer engagement.

Promotional activity remained a significant predictor of purchase amount even after controlling for income and website activity. This suggests that promotional offers can encourage customers to spend more, but their effectiveness is reinforced when customers are already actively engaging with the brand. Rather than acting independently, promotions appear to complement customer engagement by encouraging higher transaction values. From a marketing perspective, this finding highlights the importance of combining promotional campaigns with strategies that keep customers engaged throughout the purchasing process.

Overall, the final model suggests that customer spending is influenced by a combination of purchasing power, online engagement, and promotional activity. Each factor contributes in a different way: income reflects customers' spending capacity, website engagement indicates their level of interest during the purchasing journey, and promotions help increase transaction value. Together, these findings suggest that effective marketing strategies should integrate customer segmentation, digital engagement, and well-targeted promotional campaigns rather than relying on a single approach (see Appendix A).

3. Marketing Implications

Prioritize high-value customer segments.

Since income was the strongest predictor of purchase amount, marketing campaigns should be tailored to different customer segments rather than using the same strategy for everyone. Customers with greater purchasing power may respond better to premium offers, personalized recommendations, or exclusive promotions, while other segments may require different pricing or communication strategies.

Increase customer engagement before focusing on promotions.

The results showed that customers who visited the website more often and viewed more pages tended to spend more. This suggests that marketing efforts should first encourage customers to interact with the website through relevant content, personalized product recommendations, or an

improved browsing experience. Promotions are likely to be more effective when customers are already engaged with the brand.

Use promotions strategically to increase purchase value.

Promotional campaigns had a positive association with purchase amount, indicating that discounts and special offers can encourage higher spending. However, the final model also showed that promotions are only one part of the purchasing process. Instead of relying on frequent discounts, companies should use promotions to complement customer engagement strategies and maximize the value of purchases.

Combine demographic and behavioural data when designing campaigns.

The final model shows that customer spending cannot be explained by a single characteristic. Income, website activity, and promotional response all contributed to purchase amount, suggesting that marketing decisions should be based on a combination of customer attributes and behavioural data. Since the dataset also includes variables such as age, region, email opens, clicks, ad impressions, and time spent on the website, companies have the opportunity to develop richer customer profiles and design more personalized campaigns. Integrating these sources of information can improve targeting accuracy and help allocate marketing resources more effectively.

4. Limitations

Although the final regression model performed well and satisfied the main regression assumptions, some limitations should be considered when interpreting the results.

Data quality.

The dataset contained no missing values, and the exploratory analysis did not identify data quality issues that required extensive preprocessing. While a small number of potentially influential observations were detected through Cook's Distance, they represented only a small proportion of the sample and were retained because there was no evidence that they resulted from data entry

errors or had an undue influence on the overall model. Overall, the dataset provided a reliable basis for the regression analysis.

Model assumptions.

The diagnostic tests indicated that the assumptions of linearity, normality of residuals, homoscedasticity, and multicollinearity were satisfied (see Appendix B). These results increase confidence in the validity of the estimated coefficients and statistical inference. However, meeting these assumptions does not guarantee that the model captures all aspects of customer purchasing behaviour.

Model limitations.

Multiple linear regression assumes that the relationship between the predictors and the response variable is linear. Although this assumption was appropriate for the analyzed dataset, real customer behaviour is often influenced by more complex interactions and nonlinear patterns. In addition, the model explains associations rather than causal relationships, meaning that significant predictors should not be interpreted as direct causes of higher spending.

Practical interpretability.

The findings should be interpreted within the context of the analyzed dataset. Customer purchasing behaviour may vary across industries, product categories, and economic environments. As a result, the importance of variables such as income or promotional activity may differ in other business settings, and the conclusions should not be generalized without considering the characteristics of the target market.

5. Future Improvements

Collect richer customer and transactional data.

The current dataset provided enough information to build a reliable regression model, but additional customer and transactional data could further improve future analyses. Variables such as purchase frequency, average order value, customer lifetime value (CLV), loyalty program

membership, acquisition channel, and previous purchase history would provide a more complete view of customer behaviour and improve predictive accuracy. Collecting this information would also enable the company to address a broader range of business questions. For example, future analyses could focus on predicting customer churn, identifying high-value customers, or estimating future customer value, allowing marketing strategies to become more proactive and personalized.

Explore alternative modeling approaches.

Multiple linear regression proved to be an appropriate choice for this analysis because the relationships observed in the exploratory analysis were predominantly linear. However, customer purchasing behaviour is often more complex than a linear model can capture. Future analyses could compare tree-based models, such as Random Forest or Gradient Boosting, to identify nonlinear relationships and interactions between variables. Generalized Additive Models (GAMs) could also provide greater flexibility while maintaining model interpretability.

Develop customer segments before prediction.

Rather than applying a single predictive model to the entire customer base, future analyses could first group customers according to their demographic and behavioural characteristics. Segmentation techniques, such as clustering, may reveal groups with different purchasing patterns and allow organizations to build more targeted predictive models. This approach could improve marketing personalization and increase the effectiveness of future campaigns.

6. Conclusion

This project demonstrated how multiple linear regression can be used to transform customer data into meaningful marketing insights. Rather than identifying a single factor that explains customer spending, the results showed that purchase amount is associated with a combination of purchasing power, online engagement, and promotional activity. Together, these findings reinforce the importance of understanding customer behaviour from multiple perspectives instead of relying on isolated metrics.

The final model explained a substantial proportion of the variation in purchase amount while satisfying the main assumptions of multiple linear regression. Although every model has limitations, the diagnostic results indicate that the estimated relationships are statistically reliable and provide a solid foundation for interpretation and decision-making.

More importantly, this project illustrates the value of combining statistical analysis with business understanding. Regression models are not only useful for explaining past customer behaviour but also for supporting practical marketing decisions, such as customer segmentation, campaign planning, and future predictive initiatives. As organizations continue to collect richer customer and transactional data, these models can evolve into even more powerful tools for improving marketing performance.

Appendix A. Final Regression Results

Table A1. Final Regression Model Coefficients

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	-0.014	0.758	-0.018	0.985
Income	0.000503	0.000006	88.729	<0.001
Website Visits	2.002	0.045	44.295	<0.001
Pages Viewed	1.480	0.026	56.492	<0.001
Promotion Used	20.010	0.287	69.757	<0.001

Table A2. Final Model Performance

Metric	Value
Multiple R ²	0.7769
Adjusted R ²	0.7767
Residual Standard Error	10.13
F-statistic	4349
Model p-value	<0.001
AIC	37,354.76
BIC	37,393.87

Appendix B. Regression Diagnostic Tests

Table B1. Summary of Diagnostic Tests

Assumption	Test	Result	Conclusion
Linearity	Residual vs. Fitted Plot; Component + Residual Plots	Visual inspection	Linear relationship supported
Normality of residuals	Shapiro–Wilk	W = 0.9997, p = 0.7172	Assumption satisfied
Homoscedasticity	Breusch–Pagan	BP = 2.3799, p = 0.6663	No evidence of heteroscedasticity
Multicollinearity	Variance Inflation Factor (VIF)	All VIF $\approx$ 1.00	No multicollinearity detected
Influential observations	Cook's Distance	Maximum = 0.0038	No influential observations

Figure B1. Residuals vs. Fitted Values

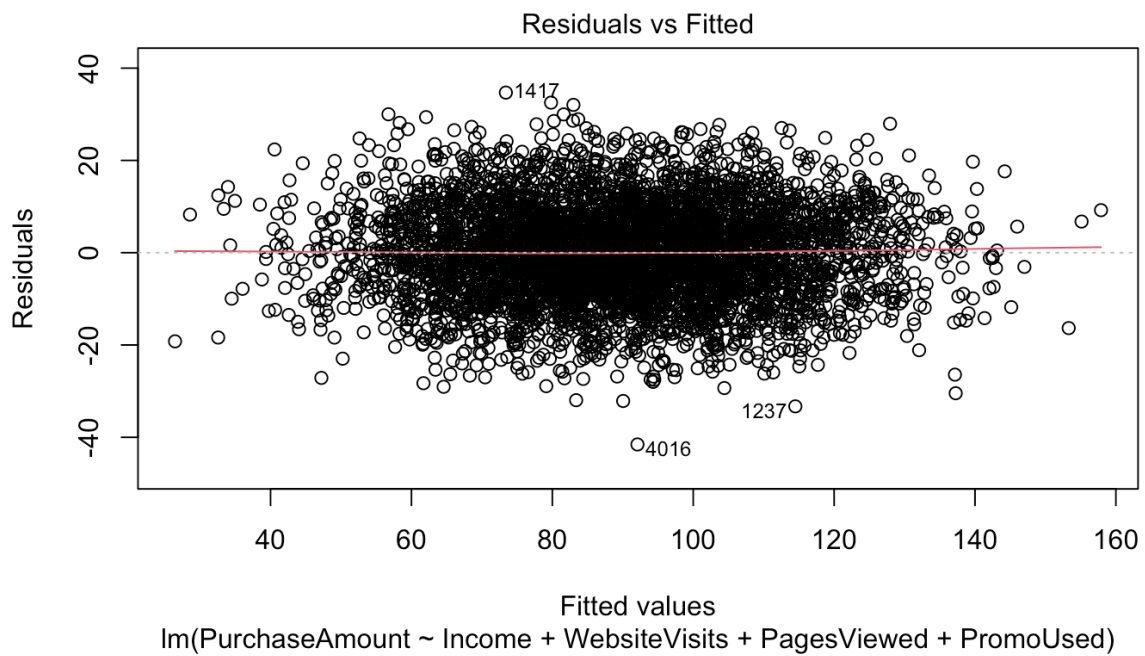


Figure B2. Normal Q-Q Plot

